

AP CSP | Chapter 4: Study Session Analyzer

Part A | Define, plan and implement

Name: _____ Partner: _____ Date: _____

Purpose: summarize goal-reaching study sessions. Use fictional data. The core input is a nonempty list of whole-number minutes and a whole-number goal, all from 0 through 180. Validation is not required.

Required behavior

Write countGoals(sessions, goal). Traverse every session, count each minutes >= goal, and RETURN the count. The caller displays that count, then "On track" if count >= 2, otherwise "Try again". The feedback threshold is a count of two, not a percentage.

1. Define the data and parameters. Explain their roles in this program.

Item	Meaning / role
sessions list	
goal parameter	
returned count	

2. Plan where the counter is initialized, how each item is tested, and when the procedure returns.

3. Sketch the procedure and its call, or record your saved code location. Attach the complete program. Include list traversal, selection, a parameterized procedure and RETURN.

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Part B | Predict, trace, test and revise

4. Predict before running. Then record the actual count and final message. Reset each test independently.

sessions / goal	Expected count / message	Actual count / message
[20,30,45,10] / 30		
[20,30,45,10] / 50		
[29,30,31] / 30		
[0,0] / 30		
Your valid test:		

5. Trace [29,30,31] with goal 30. Start the procedure counter at 0.

minutes	minutes >= goal?	counter after item
29		
30		
31		

6. Explain one benefit of the list and one benefit of the parameterized procedure.

7. Record a change, why it was needed, and a test result after the change.

8. Attribute starter code, outside resources or peer help used. Optional extension: test an empty list and label it outside the core nonempty contract.
